



AI with Rav

Learn AI the way you actually think.



DAY 5 of 30

MACHINE LEARNING

Linear Regression — your first real algorithm

WHAT YOU'LL LEARN TODAY

- › What Linear Regression actually does
- › The $y = mx + c$ formula, in plain English
- › What "best fit" means (minimising error)
- › Train one yourself in 5 lines of code

Linear Regression — your first real algorithm

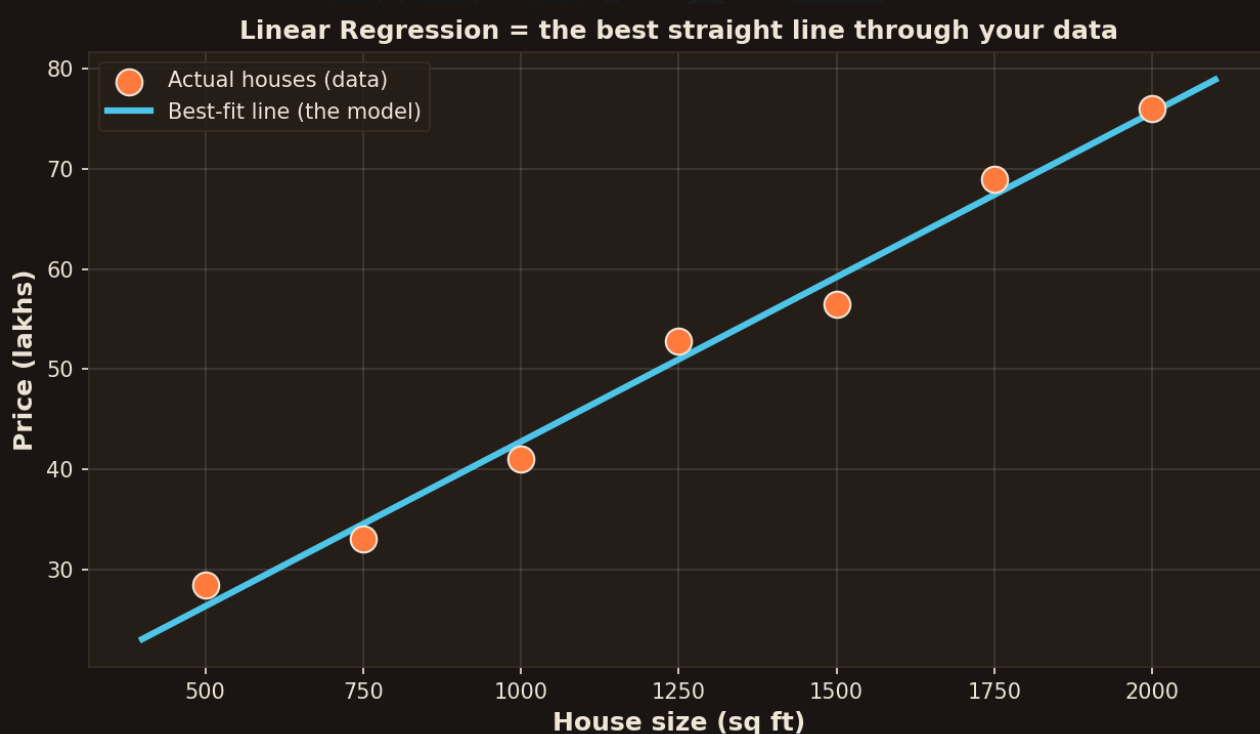
In One Line: Linear Regression draws the best straight line through your data — then uses that line to predict a number (price, sales, temperature) for anything new.

Start here: the property dealer

Ask any Mumbai property dealer the price of a 1,200 sq ft flat and they'll answer in seconds. How? Years of seeing flats: bigger → costlier. In their head, they've drawn a rough **line** between size and price.

Linear Regression is a machine doing exactly this — but with maths instead of gut feeling. It's the **simplest, most-used ML algorithm in the world**, and once you get it, every other algorithm makes more sense.

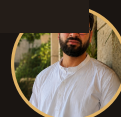
What it does: draw the best line



Each orange dot is a real house (size vs price). Linear Regression finds the one blue line that fits them best.

Give it examples — house size → price. It finds the single straight line that runs through the middle of your data. Then for a **new** size, you just read the price off the line.

Use it whenever you're predicting a **NUMBER** from inputs: house price, next month's sales, tomorrow's temperature, a student's marks. If the answer is a number → think Linear Regression first.



The formula: $y = mx + c$

The formula behind the line

$$y = m \cdot x + c$$

Diagram illustrating the components of the linear regression formula $y = mx + c$:

- y : what we predict (indicated by a red arrow)
- m : slope (how steep) (indicated by a blue arrow)
- x : the input (indicated by a red arrow)
- c : starting value (indicated by a blue arrow)

Training just finds the best m and c so the line fits your data.

Every straight line is just $y = mx + c$. Training finds the best m (slope) and c (starting point).

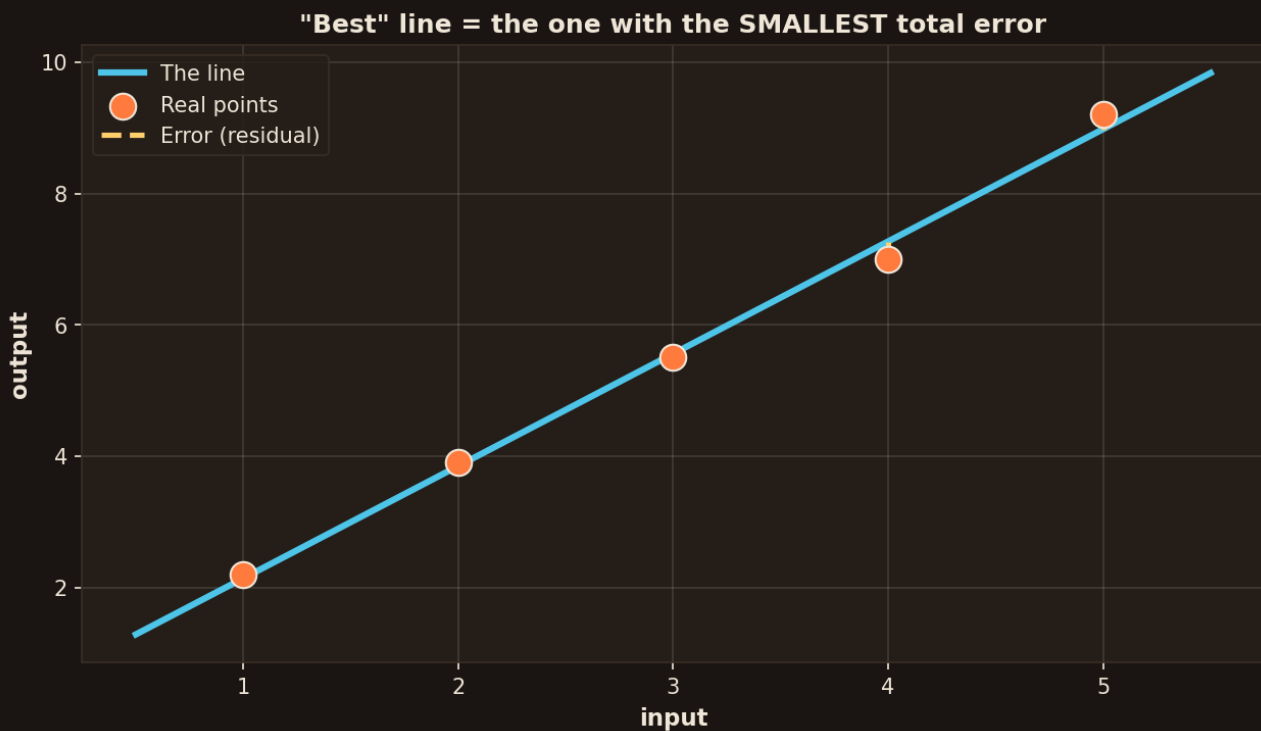
Remember school maths? A straight line is $y = mx + c$:

- y → what you predict (the price)
- m → the slope: how much price rises per extra sq ft
- x → the input (the size)
- c → the starting value (base price when size = 0)

"Training" a Linear Regression model = the machine searching for the best m and c so the line hugs your data as closely as possible. That's the whole magic — just finding two numbers.

What does "best" line mean?





The yellow dashed lines are the errors — the gap between each real point and the line. Best line = smallest total error.

There are infinite lines you could draw. The "best" one is the line where the **total distance from the points to the line is smallest**. Each gap (point to line) is an **error** (a *residual*). Add up all the errors — the best line has the least.

This "minimise the total error" idea is the seed of ALL machine learning. Every model, even ChatGPT, is just minimising an error (a "loss") — Linear Regression is where you see it in its purest form.

Train one yourself — 5 lines

```
from sklearn.linear_model import LinearRegression

X = [[500], [750], [1000], [1500], [2000]] # house sizes
y = [25, 34, 41, 58, 76] # prices (lakhs)

model = LinearRegression()
model.fit(X, y) # finds best m and c

print(model.predict([[1200]])) # new flat -> ~48 lakhs
print(model.coef_, model.intercept_) # the learned m and c
```

That's a real, working ML model in 5 lines. `model.coef_` is your **m**, `model.intercept_` is your **c** — the machine found them for you.



Recap — the 20-second version

- Linear Regression predicts a **number** by drawing the best straight line through data.
- The line is $y = mx + c$ — training just finds the best **m** and **c**.
- "Best" line = the one with the **smallest total error** (residuals).
- Minimising error is the core idea behind ALL of ML.
- 5 lines of code trains a real one.

Next up — Video 6: Logistic Regression. When the answer isn't a number but a YES/NO (spam or not? pass or fail?) — the spam-filter brain. See you Day 6.

